

Project Report on:

**Feynman's Path Integral
Formulation of Quantum Mechanics**

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Chapter 1

Introduction

This project was done as part of the course Quantum Mechanics 2, offered at CMI during January-April 2026.

This project aims understanding of Feynman's path integral formulation of quantum mechanics. The project doesn't follow a complete mathematical rigor as the aim of this project is to understand the concept and learn to use it appropriately. I start the project by stating the concept of path integral recipe and applying it to a free particle system. Then I will include a derivation of the recipe from which it will be clear what the recipe actually is. This derivation is not a complete mathematical derivation, and has some assumptions made without full justifications.

In chapter 3, I will state and prove an important result regarding some specific class of potentials which connects them to the classical action. I use this result to workout the example of harmonic oscillator.

In chapter 4, I do path integration of particle in a box in phase space coordinates.

In chapter 5, I study imaginary time formalism and its relation to statistical mechanics.

The main reference material I used for the study of this subject is Principles of Quantum Mechanics by R. Shankar [\[1\]](#).

Chapter 2

Path Integrals: Recipe & Derivation

The Schrödinger formulation of QM is based on Hamiltonian mechanics, where the path integral formulation is based on Lagrangian mechanics. We solved the quantum problem in Schrödinger formulation by finding the eigenvalues and eigenfunctions of Hamiltonian H and then finding the propagator $U(t)$ from Schrödinger equation ,

$$U(t) = e^{-\frac{iHt}{\hbar}} = \sum_n e^{-\frac{iE_n t}{\hbar}} |n\rangle \langle n|$$

where $H|n\rangle = E_n|n\rangle$.

We will discuss the path integral formulation for 1 spatial dimension, the generalization is direct.

2.1 The Recipe

In path integral formulation, we directly compute the propagator $U(x, t; x', t')$ (for propagation from (x', t') to (x, t)) by,

Path Integral Recipe

$$U(x, t; x', t') = A \sum_{\text{all paths}} e^{\frac{iS[x(t)]}{\hbar}}$$

where the summation is over all paths $x(t)$ in the $x - t$ plane from (x', t') to (x, t) . $S[x(t)]$ is the action for the path $x(t)$ and A is a normalization factor included to ensure unitarity of U .

2.2 Free Particle Propagator

We wish to compute the propagator $U(x_N, t_N; x_0, t_0)$ for a free particle (in 1 dimension). We slice the time period $t_N - t_0$ into N slices of length $\varepsilon = \frac{t_N - t_0}{N}$. We approximate the path $x(t)$ by the piecewise straight line path with straight lines between each x_{n-1} and x_n , where $x_n := x(t_n)$, $t_n := t_0 + n\varepsilon$.

We then perform the path integral,

$$\int_{x_0}^{x_N} e^{iS[x(t)]/\hbar} \mathcal{D}[x(t)]$$

where $\int_{x_0}^{x_N} \mathcal{D}[x(t)]$ is a symbol for doing integration over all paths between x_0 & x_N . The form of this integral will be clear once we go through the derivation in next section.

After, we take the limit $N \rightarrow \infty$ ($\varepsilon \rightarrow 0$) and we hope to get an answer independent of these approximations. The hope makes sense as the Lagrangian doesn't depends on acceleration and higher derivatives.

We also discretize the action

$$S = \int_{t_0}^{t_N} \frac{1}{2} m \dot{x}^2 dt$$

to

$$S = \sum_{i=0}^{N-1} \frac{m}{2} \left(\frac{x_{i+1} - x_i}{\varepsilon} \right)^2 \varepsilon.$$

Now we will start the computations. We need to calculate

$$U(x_N, t_N; x_0, t_0) = \lim_{N \rightarrow \infty} A' \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} \exp \left[- \sum_{i=0}^{N-1} \frac{(y_{i+1} - y_i)^2}{i} \right] dy_1 \cdots dy_{N-1}$$

where we have done a substitution $y_i = x_i \sqrt{\frac{m}{2\hbar\varepsilon}}$, $A' = A \left(\frac{2\hbar\varepsilon}{m} \right)^{(N-1)/2}$. This nested integral has a recursive pattern, which let us compute it,

$$\int_{-\infty}^{\infty} \exp \left[- \frac{1}{i} \left[(y_2 - y_1)^2 + (y_1 - y_0)^2 \right] \right] dy_1 = \left(\frac{i\pi}{2} \right)^{1/2} e^{-(y_2 - y_0)^2/2i}$$

$$\left(\frac{i\pi}{2} \right)^{1/2} \int_{-\infty}^{\infty} e^{-(y_3 - y_2)^2/i} e^{-(y_2 - y_0)^2/2i} dy_2 = \left[\frac{(i\pi)^2}{3} \right]^{1/2} e^{-(y_3 - y_0)^2/3i}$$

Hence all the integrals compute to $\frac{(i\pi)^{(N-1)/2}}{N^{1/2}} e^{-(y_N - y_0)^2/Ni}$.

Substituting the old variables gives

$$U(x_N, t_N; x_0, t_0) = \lim_{\substack{\varepsilon \rightarrow 0 \\ N \rightarrow \infty}} A \left(\frac{2\pi\hbar\varepsilon i}{m} \right)^{N/2} \left(\frac{m}{2\pi\hbar i N \varepsilon} \right)^{1/2} \exp \left[\frac{im(x_N - x_0)^2}{2\hbar N \varepsilon} \right]$$

Now regarding A , we need

$$\lim_{t_N - t_0 \rightarrow 0} U(x_N, t_N; x_0, t_0) = \delta(x_N - x_0) = \lim_{\Delta \rightarrow 0} \frac{1}{(\pi\Delta^2)^{1/2}} \exp \left[- \frac{(x_N - x_0)^2}{\Delta^2} \right]$$

and hence

$$A = \left(\frac{m}{2\pi\hbar\varepsilon i} \right)^{N/2}.$$

Hence we get,

$$U(x_N, t_N; x_0, t_0) = \sqrt{\frac{m}{2\pi\hbar i(t_N - t_0)}} \exp \left[\frac{im(x_N - x_0)^2}{2\hbar(t_N - t_0)} \right]$$

which is the result known from Schrödinger formulation.

Hence we note for later use that the integration over all paths is

$$\int \mathcal{D}[x(t)] = \lim_{\substack{\varepsilon \rightarrow 0 \\ N \rightarrow \infty}} \frac{1}{B} \int_{-\infty}^{\infty} \int \cdots \int_{-\infty}^{\infty} \frac{dx_1}{B} \cdot \frac{dx_2}{B} \cdots \frac{dx_{N-1}}{B}$$

where $B = \sqrt{\frac{2\pi\hbar\varepsilon i}{m}}$.

2.3 Derivation

Consider a time independent Hamiltonian $H = \frac{P^2}{2m} + V(X)$.

$$U(x, x', t) := U(x, t; x', 0) = \langle x | e^{-\frac{i}{\hbar} H t} | x' \rangle$$

We write $\exp\left(-\frac{i}{\hbar} H t\right) = \left[\exp\left(-\frac{i}{\hbar} H \frac{t}{N}\right)\right]^N$. We will work up to 1^{st} order in $\varepsilon = \frac{t}{N}$.

Now we make an assumption that up to 1^{st} order in ε ,

$$\exp\left(-\frac{i\varepsilon}{\hbar}(P^2/2m + V(X))\right) = \exp\left(-\frac{i\varepsilon}{2m\hbar}P^2\right) \exp\left(-\frac{i\varepsilon}{\hbar}V(X)\right)$$

We use the identity $e^A e^B = e^{A+B+[A,B]/2+\cdots}$ as a justification for this as the commutators and other terms following are of higher order.¹

By this assumption, we can write

$$U(x, x', t) = \langle x | \underbrace{\exp\left(-\frac{i\varepsilon}{2m\hbar}P^2\right) \exp\left(-\frac{i\varepsilon}{\hbar}V(X)\right) \exp\left(-\frac{i\varepsilon}{2m\hbar}P^2\right) \exp\left(-\frac{i\varepsilon}{\hbar}V(X)\right) \cdots}_{N \text{ times}} | x' \rangle$$

We use the completeness relation $I = \int_{-\infty}^{\infty} dx |x\rangle \langle x|$ to replace each $\exp\left(-\frac{i\varepsilon}{\hbar}V(X)\right) \exp\left(-\frac{i\varepsilon}{2m\hbar}P^2\right)$ by

$$\int_{-\infty}^{\infty} dx_n \exp\left(-\frac{i\varepsilon}{\hbar}V(X)\right) |x_n\rangle \langle x_n| \exp\left(-\frac{i\varepsilon}{2m\hbar}P^2\right)$$

As V is made up of just X operators, $\exp\left(-\frac{i\varepsilon}{\hbar}V(X)\right) |x_n\rangle = \exp\left(-\frac{i\varepsilon}{\hbar}V(x_n)\right) |x_n\rangle$.

Hence, $\langle x_{n+1} | \exp\left(-\frac{i\varepsilon}{2m\hbar}P^2\right) \exp\left(-\frac{i\varepsilon}{\hbar}V(X)\right) |x_n\rangle = \langle x_{n+1} | \exp\left(-\frac{i\varepsilon}{2m\hbar}P^2\right) |x_n\rangle \exp\left(-\frac{i\varepsilon}{\hbar}V(x_n)\right)$.

And the first term is $\langle x_{n+1} | \exp\left(-\frac{i\varepsilon}{2m\hbar}P^2\right) |x_n\rangle = \langle x_{n+1} | \int dp |p\rangle \langle p| \exp\left(-\frac{i\varepsilon}{2m\hbar}P^2\right) |x_n\rangle$

$$= \frac{1}{2\pi\hbar} \int dp \exp\left(\frac{ipx_{n+1}}{\hbar}\right) \exp\left(\frac{-ip^2}{2m\hbar}\right) \exp\left(\frac{-ipx_n}{\hbar}\right) = \left[\frac{m}{2\pi i\hbar\varepsilon}\right]^{1/2} \exp\left[\frac{im(x_{n+1}-x_n)^2}{2\hbar\varepsilon}\right]$$

Substituting this, we get

¹This is not a rigorous argument for Hilbert spaces of infinite dimension.

Path Integral

$$U(x_N, x_0, t) = \frac{1}{B} \int \exp \left[\sum_{n=1}^N \frac{i\varepsilon}{\hbar} \left(\frac{m(x_n - x_{n-1})^2}{2\varepsilon^2} - V(x_{n-1}) \right) \right] \prod_{n=1}^{N-1} \frac{dx_n}{B}$$

where $B = \sqrt{\frac{2\pi\hbar\varepsilon i}{m}}$.

Note that the exponential term is exactly the discretized version of $e^{iS/\hbar}$.

It is actually the limit of this discrete version which we denote by continuum notation $\int \mathcal{D}[x(t)] e^{iS[x(t)]/\hbar}$ where

$$\int \mathcal{D}[x(t)] := \lim_{N \rightarrow \infty} \left(\frac{m}{2\pi i \hbar \varepsilon} \right)^{1/2} \int \prod_{n=1}^{N-1} \left(\frac{m}{2\pi i \hbar \varepsilon} \right)^{1/2} dx_n$$

Chapter 3

Potentials of the form

$$V = a + bx + cx^2 + d\dot{x} + ex\dot{x}$$

In this chapter, we prove a beautiful result and look at an example of harmonic oscillator.

3.1 Propagator in terms of classical action

For potentials of the form $V = a + bx + cx^2 + d\dot{x} + ex\dot{x}$, the propagator takes the form

$$U(t_N) = A(t_N)e^{iS_{cl}/\hbar}$$

where S_{cl} is the action for the classical path.

Note that *harmonic oscillator* is of this form.

We separate the classical path $x_{cl}(t)$ from $x(t)$ by defining $y = x - x_{cl}$. We slice the time and path as before into N parts. At each t_i , we just have to vary y_i to sum through all paths, hence $dx_i = dy_i$. As all paths have same end points, we have $y(0) = 0 = y(t_N)$ and hence we have to integrate y through all loops at 0.

$$U(x, x', t_N) = \int_0 \exp \left[\frac{i}{\hbar} S[x_{cl}(t) + y(t)] \right] \mathcal{D}[y(t)]$$

where $\int_0$ denotes that the integration is over loops at 0.

$$\mathcal{L} = \frac{1}{2}m\dot{x}^2 - a - bx - cx^2 - d\dot{x} - ex\dot{x}$$

$$\begin{aligned} S[x_{cl} + y] &= \int_0^{t_N} \mathcal{L}(x_{cl} + y, \dot{x}_{cl} + \dot{y}) dt \\ &= \int_0^{t_N} \left[\mathcal{L}(x_{cl}, \dot{x}_{cl}) + \left(\frac{\partial \mathcal{L}}{\partial x} \Big|_{x_{cl}} y + \frac{\partial \mathcal{L}}{\partial \dot{x}} \Big|_{x_{cl}} \dot{y} \right) + \frac{1}{2} \left(\frac{\partial^2 \mathcal{L}}{\partial x^2} \Big|_{x_{cl}} y^2 + 2 \frac{\partial^2 \mathcal{L}}{\partial x \partial \dot{x}} \Big|_{x_{cl}} y\dot{y} + \frac{\partial^2 \mathcal{L}}{\partial \dot{x}^2} \Big|_{x_{cl}} \dot{y}^2 \right) \right] dt \end{aligned}$$

Higher derivatives of $\mathcal{L}$ vanishes.

Lets analyze this integral. The first term is just the classical action S_{cl} . The second term is integral of

$$\left. \frac{\partial \mathcal{L}}{\partial x} \right|_{x_{cl}} y + \left. \frac{\partial \mathcal{L}}{\partial \dot{x}} \right|_{x_{cl}} \dot{y} = \frac{d}{dt} \left(\left. \frac{\partial \mathcal{L}}{\partial \dot{x}} \right|_{x_{cl}} y + \left. \frac{\partial \mathcal{L}}{\partial \dot{x}} \right|_{x_{cl}} \dot{y} \right) = \frac{d}{dt} \left(\left. \frac{\partial \mathcal{L}}{\partial \dot{x}} \right|_{x_{cl}} y \right)$$

and vanishes as $y = 0$ at the endpoints.

Now focusing on the third term,

$$\frac{1}{2} \frac{\partial^2 \mathcal{L}}{\partial x^2} = -c, \quad \frac{\partial^2 \mathcal{L}}{\partial x \partial \dot{x}} = -e, \quad \frac{1}{2} \frac{\partial^2 \mathcal{L}}{\partial \dot{x}^2} = m.$$

Hence we get,

$$U(x, t; x') = e^{iS_{cl}/\hbar} \int_0^{t_N} \mathcal{D}[y(t)] \exp \left[\frac{i}{\hbar} \int_0^{t_N} \left(\frac{1}{2} m \dot{y}^2 - cy^2 - ey\dot{y} \right) dt \right]$$

which completes the proof.

3.2 Harmonic Oscillator

For the harmonic oscillator, we have $c = \omega^2$ and all other coefficients zero. As y varies through loops at 0, we can write Fourier series for y as

$$y(t) = \sum_{n=1}^{\infty} a_n \sin \frac{n\pi t}{t_N}$$

We need to compute

$$A(t_N) = \int_0^{t_N} \mathcal{D}[y(t)] \exp \left[i/\hbar \int_0^{t_N} \frac{1}{2} m (\dot{y}^2 - \omega^2 y^2) \right] dt$$

Substituting the Fourier series gives,^[2]

$$A(t_N) \propto \frac{1}{B} \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} \exp \left[\frac{im}{2\hbar} \frac{t_N}{2} \sum_{n=1}^N \left[\left(\frac{n\pi}{t_N} \right)^2 - \omega^2 \right] a_n^2 \right] \frac{da_1}{B} \cdots \frac{da_N}{B}$$

where the proportionality constant is independent of ω .

The integration can be done separately for each a_n and upon taking product, we get the ω dependent term as $\prod_{n=1}^N \left(1 - \frac{\omega^2 t_N^2}{n^2 \pi^2} \right)^{-1/2}$. Taking the limit $N \rightarrow \infty$, we get $\sqrt{\frac{\sin \omega t_N}{\omega t_N}}$. As $\omega = 0$ corresponds to the free particle $A_{\omega=0}(t_N) = \sqrt{\frac{m}{2\pi \hbar i t_N}}$, we get

$$A(t_N) = \sqrt{\frac{m\omega}{2\pi \hbar i \sin \omega t_N}}$$

Chapter 4

Particle in a Box

Now we look at the standard particle in a (1D) box^[3]. The form of potential make it a little non trivial to do the path integral. The box is from 0 to L .

We have the boundary conditions (treated as constraints here)

$$U(x'', 0, t) = U(x'', L, t) = U(0, x', t) = U(L, x', t) = 0$$

The phase space path integral takes the form ¹

$$U(q'', q', t) = \int \mathcal{D}[q] \mathcal{D}[p] \exp \left[\frac{i}{\hbar} \sum_{j=1}^N [p \Delta q - \varepsilon H(p, q)] \right]$$

Now we will find the possible paths. As reflection at the walls can happen, we group the paths to different (homotopy) classes.

First we split the group using an integer index ν called the winding number. This accounts for ν 'round trips' in the box, a distance of $2\nu L$ after reflecting at both walls ν times each. We vary ν through $\mathbb{Z}$, with opposite sign denoting opposite direction.

Now for each ν , we have terms corresponding to odd and even reflections (without 'round trip') (after rearrangement ²)

$$C_\nu(\Delta q) = \frac{1}{2} (\delta(\Delta q - q'' + q' + 2\nu L) + \delta(\Delta q + q'' - q' + 2\nu L) - \delta(\Delta q - q'' - q' + 2\nu L) - \delta(\Delta q + q'' + q' + 2\nu L))$$

(note that this also corresponds to the constraints)

For small t , we get,

$$U(q'', q', t) = (2\pi\hbar)^{-1} \sum_{\nu=-\infty}^{\infty} \iint \exp \left[\frac{i}{\hbar} \left[p \Delta q - \frac{tp^2}{2m} \right] \right] C_\nu(\Delta q) d(\Delta q) dp$$

Now doing the integrals and using the Poisson formula $\sum_{\nu=-\infty}^{\infty} e^{i\nu\theta} = 2\pi \sum_{n=-\infty}^{\infty} \delta(\theta + 2\pi n)$, we get

$$U(q'', q', t) = \sum_{n=-\infty}^{\infty} \phi_n^*(q') \phi_n(q'') e^{-\frac{i}{\hbar} E_n t}$$

where $\phi_n(q) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi q}{L}\right)$, $E_n = \frac{n^2 \pi^2 \hbar^2}{2mL^2}$. which is the known propagator.

¹This can also be done in our known configuration space path integral, which indeed is a little bit more straight forward.^[4]

²A rigorous explanation of this terms is related to some properties of the fundamental group and symmetries and statistics of the path integral. Refer ^[5]

Chapter 5

Imaginary Time Formalism

The imaginary number i in the Schrödinger equation and equivalently the propagator can be absorbed to the time coordinate by defining $\tau = it$ and work in the *euclidean space* which we are familiar from relativity.

$$-\hbar \frac{d}{d\tau} |\psi\rangle = H |\psi\rangle, \quad U(\tau) = e^{-H\tau/\hbar}$$

This does not change the energy eigenvalues and eigenfunctions.
The path integrals becomes,

$$U(x, x', \tau) = \int \mathcal{D}[x(\tau')] \exp \left[-\frac{1}{\hbar} \int_0^\tau \mathcal{L}_E(x, \dot{x}) d\tau' \right]$$

where

$$\int \mathcal{D}[x(\tau')] = \lim_{N \rightarrow \infty} \left(\frac{m}{2\pi\hbar\varepsilon} \right)^{1/2} \prod_0^{N-1} \left(\frac{m}{2\pi\hbar\varepsilon} \right)^{1/2} dx_i, \quad \varepsilon = \frac{\tau}{N}$$

and the *euclidean Lagrangian*

$$\mathcal{L}_E = \frac{m}{2} \left(\frac{dx}{d\tau} \right)^2 + V(x)$$

(Note that there is a sign change for the real-time potential, this becomes useful in study of tunneling.)

What was a phase factor before, became a scaling factor now.

The propagator in this formalism is Hermitian instead of unitary. Thus the norms of the states are not preserved in time. Note also that after long time, every state evolves into the least energy state (l) it overlaps with.

$$\langle x | U(\tau) | x' \rangle = \sum_n \langle x | n \rangle \langle n | x' \rangle \exp \left(-\frac{1}{\hbar} E_n \tau \right) \xrightarrow{\tau \rightarrow \infty} \langle x | l \rangle \langle l | x' \rangle \exp \left(-\frac{1}{\hbar} E_l \tau \right)$$

The imaginary time formalism is used in study of many fields, we look at the example of Quantum Statistical Mechanics.

Connection to Quantum Statistical Mechanics

The quantum partition function is given by, $(\beta = \frac{1}{kT})$

$$Z = \text{Tr } e^{-\beta H} = \int_{-\infty}^{\infty} \langle x | e^{-\beta H} | x \rangle dx = \int_{-\infty}^{\infty} \langle x | e^{-\frac{1}{\hbar} \beta \hbar H} | x \rangle dx = \int_{-\infty}^{\infty} U(x, x, \beta \hbar) dx$$

Now Z has become the sum over amplitudes to go over closed paths in imaginary time $\beta \hbar$.

One important thing to notice is that taking the classical limit of this equation gives the standard $(2\pi\hbar)^{-1}$ factor in classical statistical mechanics.

Chapter 6

Summary

We looked at Feynman's path integral formulation of quantum mechanics where you directly find the propagator without getting into the eigenvalue problem. This was achieved through 'summing over all paths' with a phase associated. The major task in this formulation is finding the form of the path integral $\int \mathcal{D}[x]$ itself.

One important property of the path integrals is that how it is connected to the classical path. The contribution from paths far away from the classical path undergo destructive interference. This interference way of looking at path integrals is useful in analyzing problems like diffraction of waves.

The classical paths have more significance especially in some specific class of potentials. We used this to solve the quantum harmonic oscillator.

Then we introduced the phase space path integral which has almost the same derivation as configuration space, except you introduce more resolution of identity in terms of $|x\rangle\langle x|$ between the other exponential terms. We looked at the particle in a box problem in the phase space path integral, which indeed requires more complex methods of defining the path integral.

Finally we studied the imaginary time formulation in brief and looked at its connection with statistical mechanics. This formulation has even more important applications in many fields, especially in condensed matter. One major difference is that what was a phase before, changed to a weight-age.

Path integral formulation is a very vast and useful subject coming into study of many fields of physics. Path integrals becomes more important when studying quantum field theory. Making the path integral well defined and rigorous is another important field of study.

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